

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

A
Presentation
On

“Error detection and optimized streetlight management”

G.DHANUSH
B.SAI SUBAHMANYA

Project Supervisor
Dr. T.Sunil kumar
Dept. of E.C.E.

CONTENTS

- OBJECTIVE
- INTRODUCTION
- CHALLENGES
- EXISTING METHOD
- PROPOSED METHOD
- BLOCK DIAGRAM
- RESULTS
- ADVANTAGES
- LIMITATIONS
- APPLICATIONS
- SOFTWARE AND HARDWARE REQUIREMENTS
- REFERENCES

OBJECTIVES



**Optimize
Energy Use**



**Swift Fault
Detection**



**Enhanced
Management**





INTRODUCTION

Cloud integration

Revolutionizing street lights with a blend of photodiode, CT coils, GSM, GPS, and cloud technology.

GPS integration pinpoints errors swiftly for efficient corrective measures.

CT coils for power analysis, and GSM for instant notifications.

EXISTING METHODS



**Fixed Time
Activation**



**Manual Malfunction
Detection**



**Limited Data
Management**



CHALLENGES



The incorporation of CT coils.



sensor's dependability



Protecting sensitive information.



may need time to adapt to this advanced system.

PROPOSED METHOD



Photodiodes manage light , CT coils monitor power for swift maintenance alerts in streets.



GSM sends alerts on faults, GPS locates errors, enhancing street light system efficiency.

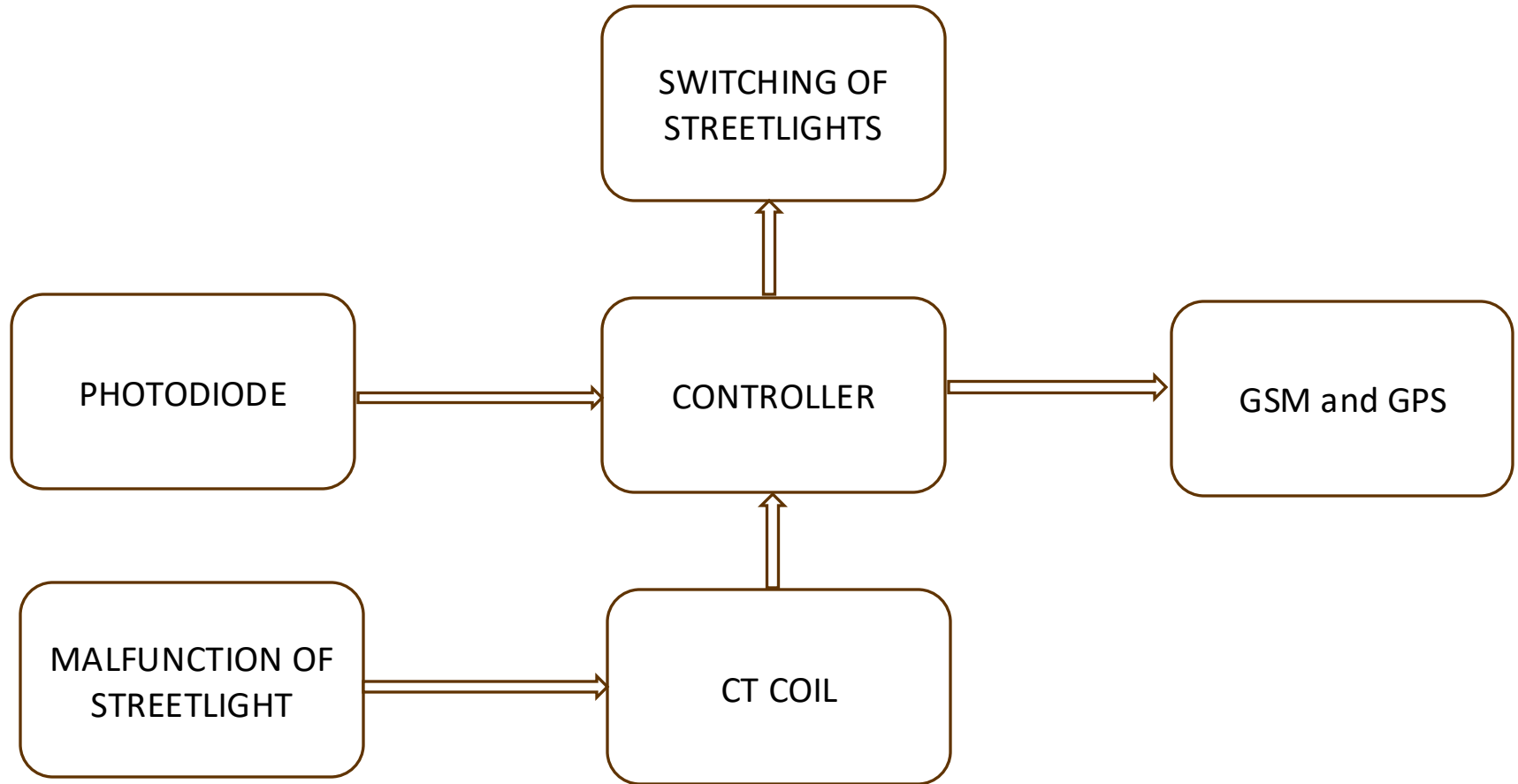


Integrates seamlessly with current street lights, enhancing functionality without altering existing infrastructure.



Cloud stores data for accessibility, enhancing reliability and optimizing street lighting.

BLOCK DIAGRAM



CODE

finalcode_miniproject.ino

```
1  #include <SoftwareSerial.h>
2  #include <EmonLib.h>
3
4  EnergyMonitor emon1; // Create an instance
5
6  int CT_pin = A0;      // CT sensor pin connected to A0 pin of Arduino
7  int relay_pin = 7;    // Changed to avoid conflict
8  int sensorPin = A1;   // Select the input pin for the photodiode
9  int ledPin = 6;       // Select the pin for the LED (changed to avoid conflict)
10 int sensorValue = 0;  // Variable to store the value coming from the sensor
11
12 #define GSM_TX 2
13 #define GSM_RX 3
14
15 SoftwareSerial gsmSerial(GSM_TX, GSM_RX);
16
17 bool callMade = false; // Track if a call has been made
18 bool messageSent = false; // Track if a message has been sent
19
20 #define THINGSPEAK_CHANNEL_ID XXXXX // Replace with your ThingSpeak channel ID
21 #define THINGSPEAK_API_KEY "XXXXXXXXXXXX" // Replace with your ThingSpeak API Key
22
23 void setup() {
24     pinMode(ledPin, OUTPUT);
25     pinMode(relay_pin, OUTPUT);
26     emon1.current(CT_pin, 32.59); // Current: input pin, calibration.
27     Serial.begin(9600);
28     gsmSerial.begin(9600);
29
30     Serial.println("Initializing GSM module...");
31     delay(2000);
32
33     gsmSerial.println("AT");
34     delay(1000);
35     if (gsmSerial.find("OK")) {
```

CODE

finalcode_miniproject.ino

```
35     if (gsmSerial.find("OK")) {
36         Serial.println("GSM module is ready.");
37     } else {
38         Serial.println("Error: GSM module not responding.");
39         while (1);
40     }
41 }
42
43 void loop() {
44     // Sensor reading
45     sensorValue = analogRead(sensorPin);
46     Serial.print("Photo diode value = ");
47     Serial.println(sensorValue);
48
49     if (sensorValue < 15) {
50         digitalWrite(ledPin, HIGH);
51         digitalWrite(relay_pin, HIGH);
52         delay(5000);
53     } else {
54         digitalWrite(ledPin, LOW);
55         digitalWrite(relay_pin, LOW);
56         delay(5000);
57     }
58
59     double Irms = emon1.calcIrms(1480);
60
61     Serial.print("POWER = ");
62     Serial.println(Irms * 230.0);
63     Serial.print("IRMS = ");
64     Serial.println(Irms);
65     Serial.print("IM = ");
66     Serial.println(Irms * 1.414);
67 }
```

CODE

finalcode_miniproject.ino

```
69   if (Irms <= 3.3 && Irms > 2.50) {
70       // Check if a call has not been made yet
71       if (!callMade) {
72           Serial.println("Making a call...");
73           gsmSerial.println("ATD+91 XXXXXXXX;"); // Replace with the desired phone number
74           delay(10000); // Wait for 10 seconds (adjust as needed)
75           gsmSerial.println("ATH"); // Hang up the call
76           callMade = true; // Mark that the call has been made
77       }
78
79       // Check if a message has not been sent yet
80       if (!messageSent) {
81           Serial.println("Sending a message...");
82           gsmSerial.println("AT+CMGF=1"); // Set SMS mode to text mode
83           delay(1000);
84           gsmSerial.println("AT+CMGS=\"+91 XXXXXXXX\""); // Replace with the recipient's phone number
85           delay(1000);
86           gsmSerial.print("!!HLO OPERATOR STREET LIGHT FAULT OCCURRED!!"); // Replace with your message
87           delay(1000);
88           gsmSerial.write(26); // Send Ctrl+Z to indicate the end of the message
89           delay(5000); // Wait for a response (adjust as needed)
90           messageSent = true; // Mark that the message has been sent
91       }
92   } else {
93       // Reset the flags to allow calling and messaging again
94       callMade = false;
95       messageSent = false;
96   }
97
98   // Push data to ThingSpeak
99   sendDataToThingSpeak(sensorValue, Irms);
100 }
```

CODE

finalcode_miniproject.ino

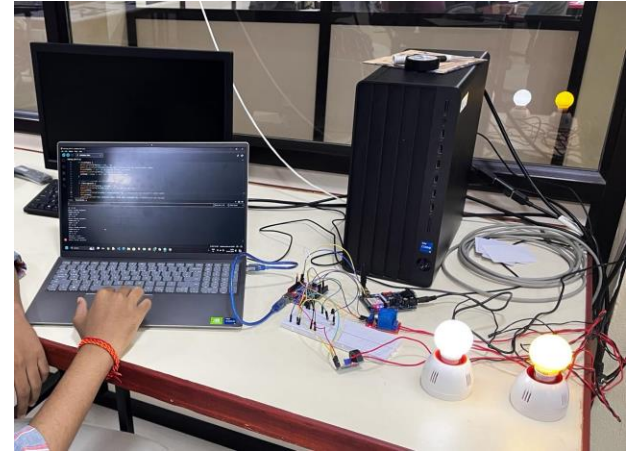
```
102 void sendDataToThingSpeak(int photodiodeValue, float IRMS) {
103     // GSM module setup
104     gsmSerial.println("AT");
105     delay(1000);
106     gsmSerial.println("AT+CGATT=1");
107     delay(1000);
108     gsmSerial.println("AT+CSTT=\"internet.ng.airtel.com\"");
109     delay(1000);
110     gsmSerial.println("AT+CIICR");
111     delay(1000);
112     gsmSerial.println("AT+CIFSR");
113     delay(3000);
114     gsmSerial.println("AT+CIPSTART=\"TCP\", \"api.thingspeak.com\", \"80\"");
115     delay(3000);
116     gsmSerial.println("AT+CIPSEND");
117     delay(2000);
118
119     // Send data to ThingSpeak
120     String postStr = String(THINGSPEAK_API_KEY);
121     postStr += "&field1=";
122     postStr += String(photodiodeValue);
123     postStr += "&field2=";
124     postStr += String(IRMS);
125     postStr += "\r\n\r\n";
126
127     gsmSerial.print("POST /update HTTP/1.1\n");
128     gsmSerial.print("Host: api.thingspeak.com\n");
129     gsmSerial.print("Connection: close\n");
130     gsmSerial.print("X-THINGSPEAKAPIKEY: ");
131     gsmSerial.print(THINGSPEAK_API_KEY);
132     gsmSerial.print("\n");
133     gsmSerial.print("Content-Type: application/x-www-form-urlencoded\n");
134     gsmSerial.print("Content-Length: ");
135     gsmSerial.print(postStr.length());
136     gsmSerial.print("\n\n");
```

CODE

```
127     gsmSerial.print("POST /update HTTP/1.1\n");
128     gsmSerial.print("Host: api.thingspeak.com\n");
129     gsmSerial.print("Connection: close\n");
130     gsmSerial.print("X-THINGSPEAKAPIKEY: ");
131     gsmSerial.print(THINGSPEAK_API_KEY);
132     gsmSerial.print("\n");
133     gsmSerial.print("Content-Type: application/x-www-form-urlencoded\n");
134     gsmSerial.print("Content-Length: ");
135     gsmSerial.print(postStr.length());
136     gsmSerial.print("\n\n");
137     gsmSerial.print(postStr);
138     gsmSerial.print((char)26);
139
140     delay(5000); // Delay before next update
141 }
142
```

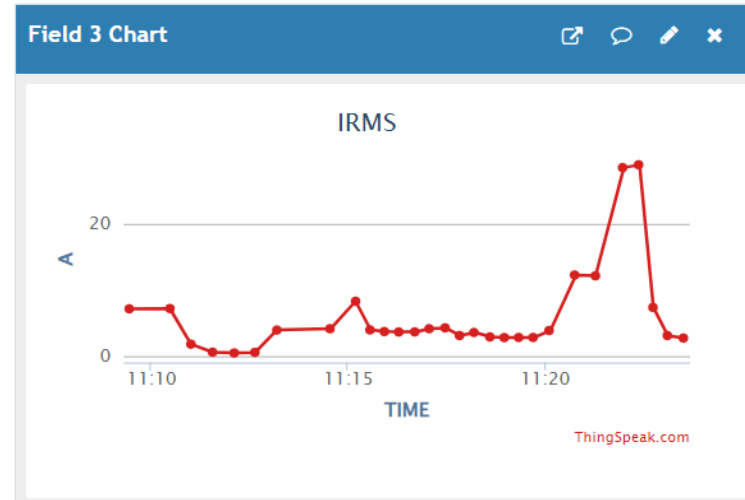
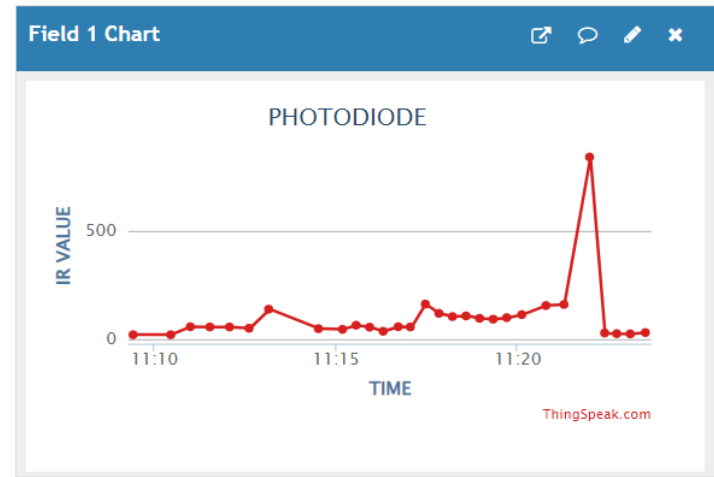
RESULT

- REDUCED ENERGY CONSUMPTION
- IMPROVED MAINTENANCE
- SUSTAINABLE SOLUTION
- DATA-DRIVEN DECISION MAKING



RESULTS

1. Centralized Data Storage
2. Real-time Accessibility
3. Enhanced Analytics



ADVANTAGES

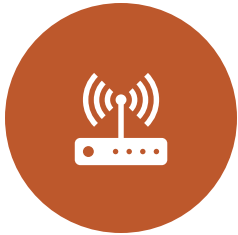
**Swift Fault
Detection**

**Energy
Efficiency**

**Precise Error
Localization**

**Effortless
Enhancement:.**

LIMITATIONS



**DEPENDENCY ON
CONNECTIVITY**



**ADAPTATION
PERIOD**



**DATA SECURITY
CONCERNS:**



**TECHNICAL
COMPLEXITY**

APPLICATIONS



**SMART CITY
INITIATIVES**



**THEFT CURRENT
DETECTION**



**PUBLIC SAFETY
ENHANCEMENT**



**ENVIRONMENTAL
CONSERVATION**

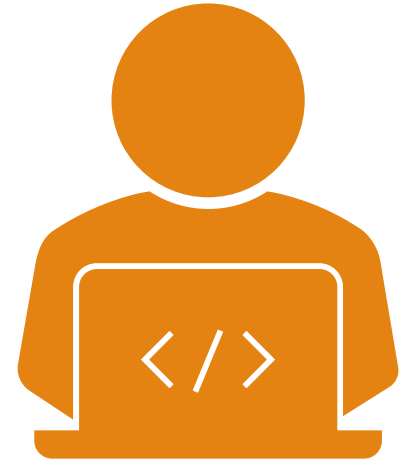
SOFTWARE AND HARDWARE REQUIREMENT

Software Requirements:

- Arduino IDE
- Thingspeak

Hardware Requirements:

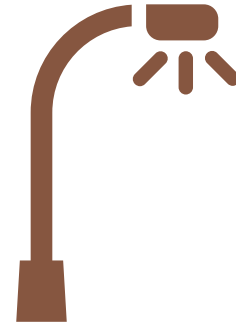
- CT Coil
- GSM and GPS Modules
- Arduino UNO
- Solid State Relay
- Photodiode



REFERENCES



José R. Celaya, José M. de la Cruz, et al. "Fault Detection and Diagnosis in Industrial Systems"



"A Review of Smart Street Lighting Control Mechanisms and Standards" by M. A. Khalaf et al. (2020) discusses various light control methods, including photodiodes.

thank you!